

Allosteric effects of the coupling cation in melibiose transporter MelB

Reviewed Preprint

v1 • September 12, 2025

Not revised

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eLife Assessment

This manuscript presents **useful** insights into the molecular basis underlying the positive cooperativity between the co-transported substrates (galactoside sugar and sodium ion) in the melibiose transporter MelB. Building on years of previous studies, this work improves on the resolution of previously published structures and reports the presence of a water molecule in the sugar binding site that would appear to be key for its recognition, introduces further structures bound to different substrates, and utilizes HDX-MS to further understand the positive cooperativity between sugar and the co-transported sodium cation. Although the experimental work is **solid**, the presentation of the data lacks clarity, and in particular, the HDX-MS data interpretation requires further explanation in both methodology and discussion, as well as a clearer description of the new insight that is obtained in relation to previous studies. The work will be of interest to biologists and biochemists working on cation-coupled symporters, which mediate the transport of a wide range of solutes across cell membranes.

<https://doi.org/10.7554/eLife.108335.1.sa4>

Abstract

The major facilitator superfamily (MFS) transporters play significant roles in human health and disease. *Salmonella enterica* serovar Typhimurium melibiose permease (MelB_{St}), which catalyzes the symport of galactosides with Na⁺, H⁺, or Li⁺, is a prototype of this important transporter superfamilies. We have published the structures of the inward- and outward-facing conformations of MelB_{St} with galactoside or Na⁺ bound, and determined the binding thermodynamic cycle. We have proposed that positive cooperativity between the two co-transported solutes plays a key role in the symport mechanism of MelB_{St}; however, the molecular basis for this core mechanism remains unclear. In this study, we investigated the structural dynamics induced by melibiose, Na⁺, or both on MelB_{St} using hydrogen-deuterium exchange mass spectrometry (HDX-MS). We also refined the specific determinants for the sugar recognition in both protein and galactoside molecules by solving the crystal structures of D59C MelB_{St} bound to melibiose and two other sugars that contain different numbers of sugar units, and identified a critical water molecule as part of the specific determinants from a α-NPG-bound structure. Our integrated structural and HDX-MS analyses support the notion

that the binding of the coupling cation at a remote site stabilizes those dynamic sidechains in the sugar-binding pocket, leading to a high-affinity state. This study provides the molecular basis for the essential symport mechanism through positive cooperativity, which may serve as a general mechanism for most cation-coupled symporters.

Introduction

The Solute Carrier (SLC) family of transporters encompasses diverse superfamilies of membrane proteins with various protein folds and employing different transport mechanisms to facilitate the translocation of a wide range of solutes across cell membranes (1). The largest superfamily of SLC transporters is the major facilitator superfamily (MFS)(2). MFS transporters comprise a significant number of cation-coupled secondary active transporters and are responsible for the uptake of a broad spectrum of solutes across cell membranes, playing crucial roles in physiology, pathology, and pharmacokinetics, and are emerging as drug targets (3, 4). Recent rapid advancements in membrane protein research have greatly enhanced our understanding of protein conformation and mechanisms(5–8); however, critical details remain lacking. For example, in cation-coupled symport, it is still unclear how the coupling cations help in the binding, translocating, and accumulating the primary substrate.

For MFS secondary active transporters, most members use H^+ as the coupling cation, and a few members use Na^+ , such as the Na^+ -coupled lipid transporter (MFSD2A)(9, 10) that is expressed in the major organ barriers, including the blood-brain barrier or blood-retina barrier. MFSD2A plays a critical role in the uptake of essential lipids into these neural tissues (9–11). The Na^+ -coupled melibiose transporter of *Salmonella enterica* serovar Typhimurium (MelB_{St}), which is a well-characterized representative for the Na^+ -coupled MFS transporters, catalyzes the symport of a galactopyranoside with Na^+ , H^+ , or Li^+ , and is a useful model system for studying cation-coupled transport mechanisms (12–22). Two major conformations of MelB_{St} have been determined: one is an outward-facing conformation at apo or galactoside-bound states(20, 22), and the other is an inward-facing conformation at a Na^+ -bound state(21). The primary substrate-specificity determinant pocket and the cation-specificity determinant pocket have been structurally and functionally characterized (19–23). Binding at both pockets exhibits a positive cooperativity; that is, the binding of one increases the affinity for the other (19). All three coupling cations compete for the same binding pocket, and the transport stoichiometry is 1 galactoside: 1 cation (Na^+ , H^+ , or Li^+). The sugar affinity depends on the cation identity, with the cooperativity numbers (fold of increase in affinity) being 8, 5, or 2 for Na^+ , Li^+ , and H^+ , respectively (19, 20). Notably, the sugar-binding affinity is also dependent on MelB_{St} conformation (21), in addition to being sensitive to the binding of the coupling cation and its identity. By trapping MelB_{St} in an inward-facing state using the inward-facing conformation-specific binder nanobody-725 (Nb725), both experimental sugar-binding assays and cryoEM structural analysis support a low-affinity state of the sugar-binding pocket (21, 23). Remarkably, Na^+ binding to the inward-facing conformation remains unchanged (21, 23). These results provide experimental evidence supporting the previously proposed stepped-binding kinetic model for melibiose/ Na^+ symport, in which Na^+ binds first and is released after the sugar release on the opposite surface (15, 18, 24, 25).

Positive cooperativity is proposed to be the key symport mechanism in MelB_{St}, but the molecular basis for this critical mechanism remains unclear. The structures indicate that the bound sugar and Na^+ have no direct contact, while the two binding pockets are in close proximity (20, 21). The minimum free-energy landscape for sugar translocation has been simulated based on the structures of the outward- and inward-facing conformations as two starting points (26). The results suggest that the Na^+ contribution to the binding free energy of sugar by direct contact is negligible. The cooperativity between melibiose and Na^+ occurs through allosteric coupling, primarily via electrostatic effects. To experimentally evaluate this concept, in this study, we

enhanced the resolution of the structure with α -NPG bound, which revealed a critical water molecule, and also determined crystal structures with three other α -sugars (melibiose, raffinose, or α -MG) that contain different numbers of sugar units to verify the sugar-recognition specific determinants. We also analyzed the structural dynamics with HDX-MS induced by melibiose, Na^+ , or both on MelB_{St}. The extensive integrated analyses of the structure and structural dynamics by MD simulations and HDX-MS support the conclusion that the binding of the coupling cation at a remote location stabilizes the sugar-binding residues to switch to a higher-affinity state. Therefore, the coupling cation in this symporter is an allosteric activator.

Results

Outward-facing structures of MelB complexed with varied sugar substrates

We have published the crystal structures of MelB_{St} with bound sugar analogues α -NPG or melibiose-derived detergent (DDMB), from a MelB_{St} uniport mutant D59C (20 [DOI](#)), or a solely H⁺-coupled symporter mutant D55C (22 [DOI](#)). Here we report four crystal structures of D59C MelB_{St} with the bound endogenous sugar substrate melibiose, and two other α -galactosides, α -methyl galactoside (α -MG) with a single sugar unit or raffinose with three sugar units, as well as α -NPG at an improved resolution. The statistics of the four structures are presented in Table S1 [DOI](#). All outward-facing structures are virtually identical with RMSD values of less than 0.4 (Fig. S1 [DOI](#)). This typical MFS-fold transporter has 12 transmembrane helices and is organized into two six-helix domains linked by the middle loop between helices VI and VII (Loop₆₋₇). There are two cytoplasmic helices on the middle loop or the C-terminal tail or lid (Fig. 1 [DOI](#)). In all structures, one sugar molecule is bound in the middle of the protein and sandwiched by the helices from both the N- and C-terminal domains. As indicated by the sliced surface, the binding residues located within the cytoplasmic leaflet of both domains shape the edge of the inner barrier that prevents the sugar from passing across the transporter into the cytoplasm. On the periplasmic side, there is an open vestibule connecting the solvent to the binding pocket.

α -NPG binding refined to a resolution of 2.7 Å

The improved resolution from 3.01 Å to 2.7 Å provides a better-resolved density map for the bound α -NPG molecule, which further supports the originally assigned pose (Fig. 2 [DOI](#)) (20 [DOI](#)). As described, the binding pocket is formed by 14 residues on five helices, including N-terminal helices I (Lys18, Asp19, Ile22, and Tyr26), IV (Tyr128, Asp124, Tyr120), and V (Arg149 and Ala152), and the C-terminal helices X (Trp342) and XI (Gln372, Thr373, Val376, and Lys377).

A water molecule was modeled to a positive density, which located at a good hydrogen-bonding distances from the C4-OH on the galactopyranosyl ring and Thr733 on helix XI, and also surrounded by C6-OH, Gln372 on helix XI and Asp124 on helix IV (Fig. 2 [DOI](#)). Overall, the galactosyl moiety is primarily stabilized by the N-terminal charged residues Asp19, Asp124, and Arg149. Especially, strong and continuous density between Asp19 and the C2-OH and C3-OH and Asp124 with C4-OH were observed in the previous maps (20 [DOI](#)) and current maps, indicating that both negatively charged Asp residues and the water molecule play critical roles in the binding. Five more water molecules were modeled, and three interact with the previously identified charge network involved in 9 charged residues between the N- and C-terminal domains (Fig. S2a, b [DOI](#)). The water 2 and 3 interact with the charged pair Arg295 (helix IX) and Glu351 (helix XI) and the water 4 interacts with Arg138 on helix IV, which also form a salt bridge with Glu142 to stabilize the hydrogen-bonding interaction between Glu142 and Arg295 (Fig. S2b [DOI](#)).

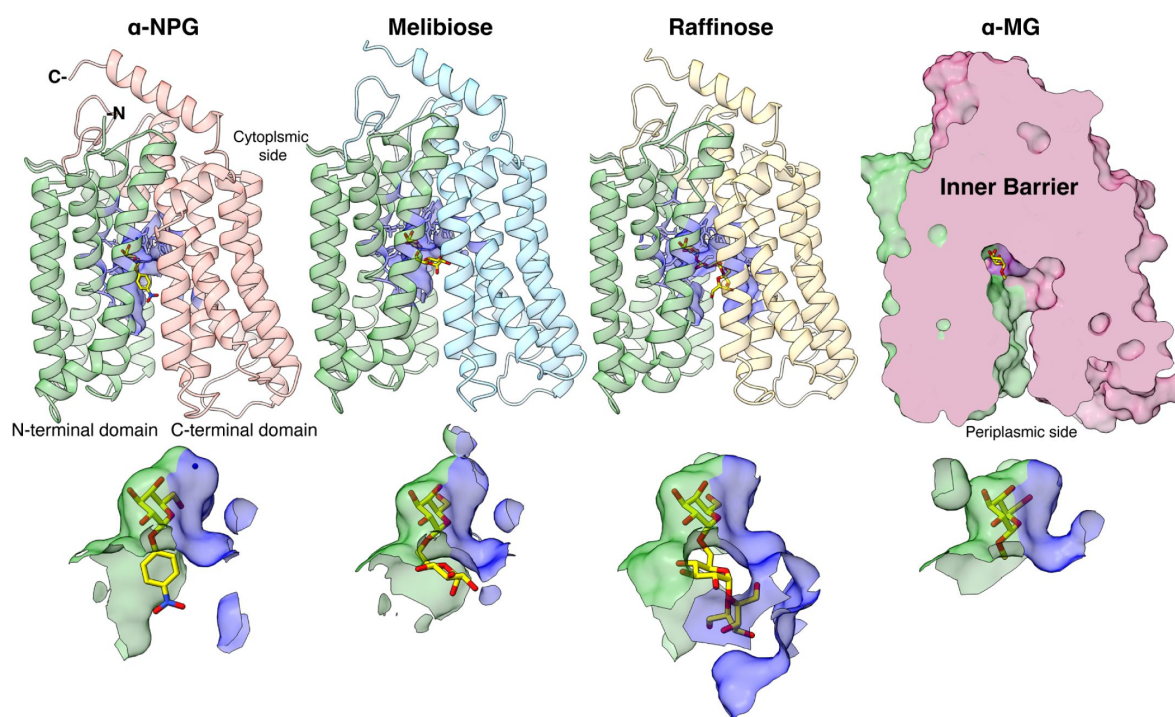


Fig. 1.

Crystal structure of D59C MelBSt in complex with α -sugars or sugar analogs.

The identities of the bound sugars are indicated at the top. Upper row: Cartoon representation of the structures of D59C MelB_{St} bound with α -nitrophenyl galactoside (α -NPG), melibiose (α -disaccharide), and raffinose (α -trisaccharide), respectively, along with a surface presentation of D59C MelB_{St} complexed with α -methyl galactoside (α -MG). All structures were oriented with the cytoplasmic side on top and the N-terminal domain (colored green) on the left. Each sugar molecule is colored yellow. The blue sticks and surfaces indicate residues within 5 Å of the sugar molecules. Lower row: the sugar-binding pockets. Residues from the N-terminal and C-terminal domains were shown in surface representation, and colored in green and blue, respectively.

Melibiose binding

The endogenous substrate melibiose is formed from a galactose unit and a glucose unit linked by an α -1,6 galactosyl bond (D-Gal-(α 1 – 6)-D-Glc). The melibiose-bound D59C MelB_{St} structure was refined to a resolution of 3.05 Å, and the density map displayed a clear two-unit blob in the binding pocket, fitting well with one molecule of melibiose. While the water molecule was not modeled due to limited resolution, a positive density is also present at W1 position. Two waters at W2 and W3 positions were also modeled in this structure. Interestingly, the binding affinity between melibiose and α -NPG differs by a factor of 100; the only difference in the binding sites is the pose and chemistry of the non-galactosyl moiety (**Fig. 3a, e**). In α -NPG, the hydrophobic phenyl group, positioned at 4 Å-distance from the phenyl group of Tyr26, forms a strong stacking interaction. While melibiose has four additional OH groups, there is no further polar interaction with MelB. This glucosyl ring rotates away and becomes nearly perpendicular to the phenyl ring of Tyr26 (**Fig. S3**). The absence of aromatic stacking may contribute to the relatively poor affinity of melibiose.

α -Methyl galactoside binding

This sugar molecule has a methyl substituent at the anomeric C1 position in an α -linkage. The α -MG-bound structure of the D59C MelB_{St} mutant was refined to a resolution of 3.68 Å, and the density map still clearly displayed a one-unit blob in the binding pocket, where one α -MG was modeled similarly to the galactosyl moiety of melibiose (**Fig. 3b**).

Raffinose binding

Raffinose is a trisaccharide formed from melibiose and fructose. The raffinose-bound D59C MelB_{St} mutant structure was refined to a resolution of 3.40 Å, and the density map displayed a clear three-unit blob in the binding pocket, fitting well with this trisaccharide. As expected, the raffinose-binding pocket is significantly larger than that of melibiose and α -NPG (**Fig. 3c**), involving two additional helices. Consequently, a set of polar side chains on helix VII (Asn244, Ser247, Asn248, and Asn251), helix VIII (Asn279), and helix V (Ser153) were found to be within 4-5 Å of either the glucosyl or fructosyl moieties.

Interestingly, the polar interactions at hydrogen-bonding distances between raffinose and MelB_{St} are nearly identical to those in α -MG, melibiose, or α -NPG, with most occurring within the galactosyl moiety, while the Asp19 is at a hydrogen-bonding distance from the OH-4 on the glucosyl moiety.

All four galactosyl moieties align well in the specificity-determinant pocket, regardless of the number of monosaccharide units (**Fig. 3d, e**; **Fig. S1**). This alignment strongly supports the previous conclusion that the galactosyl moiety determines the specificity of the primary substrates; however, while the non-galactosyl moiety contributes to the binding affinity. The galactosyl-binding pocket is well formed with little room for flexibility in the sugar pose; however, the spacious cavity can accommodate at least one more fructose unit, as shown in the raffinose.

Isothermal titration calorimetry (ITC) measurement of the binding affinity

Binding affinity K_d values of the WT and D59C MelB_{St} for melibiose of $1.25 \text{ mM} \pm 0.05 \text{ mM}$ and $4.96 \pm 0.11 \text{ mM}$, respectively, and for α -NPG of $16.46 \pm 0.21 \text{ } \mu\text{M}$ or $11.97 \pm 0.09 \text{ } \mu\text{M}$, respectively, have been determined by ITC (19, 20, 25, 27). To determine the binding affinity of MelB_{St} for α -MG and raffinose, ITC measurements were also utilized to determine the K_d values in the presence of Na⁺ (**Fig. 4**). By injecting the raffinose or α -MG in a buffer-matched solution into the

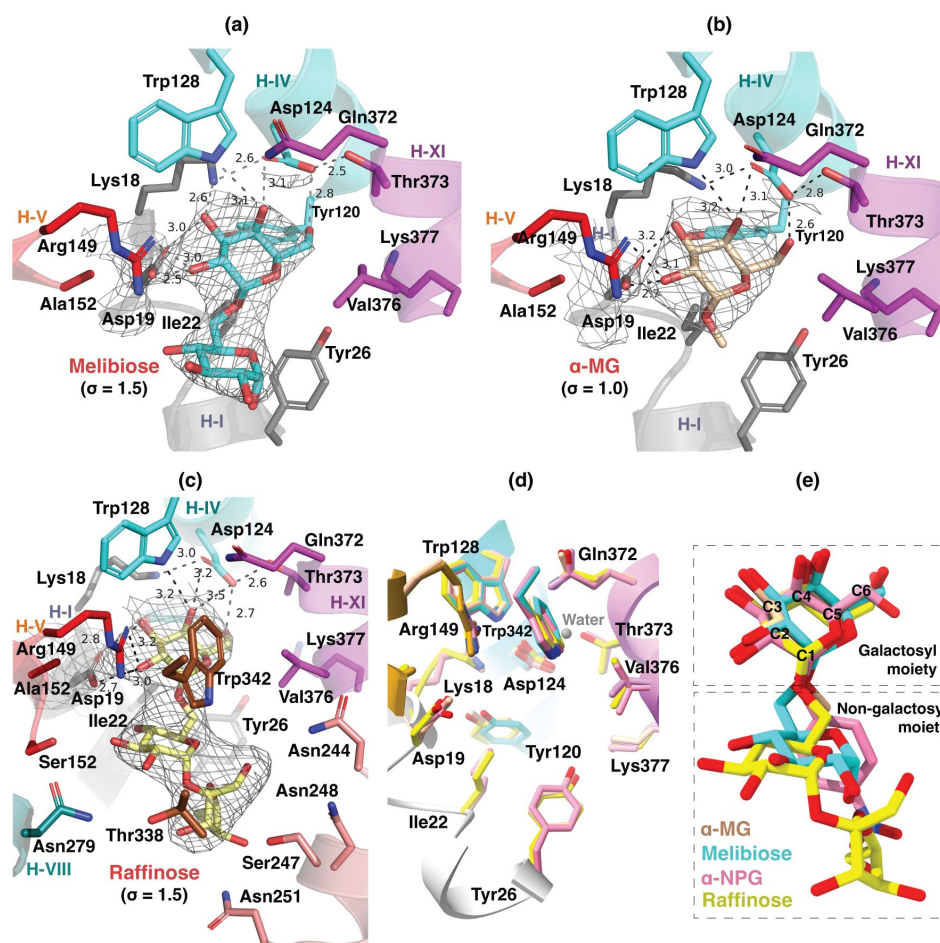


Fig. 3.

Binding of melibiose, α -MG, and raffinose.

All residues within a 5-Å distance to the bound substrates were shown in sticks. Isomesh maps for each sugar and Asp19 were contoured at levels of $\sigma = 1.5$ for melibiose and raffinose or $\sigma = 1.0$ for α -MG. Dashed lines, the distances within hydrogen-bonding and salt-bridge interactions (Å). **(a)** Melibiose binding. **(b)** α -MG binding. **(c)** Raffinose binding. **(d)** Residues in the sugar-binding pockets from the alignment of all four structures. **(e)** Substrates from the alignment of all four structures. Carbon positions on the galactosyl (C1-6) and glucosyl moiety (C1-6') are labeled on the melibiose molecule. Trp342 was removed for clarity in panels a, b, and d.

sample cell containing the WT MelB_{St} in the presence of Na⁺, the isotherm curve fitted well, yielding K_d values of 0.88 ± 0.04 mM or 3.79 ± 0.53 mM for α -MG or raffinose, respectively. It is surprising that raffinose, which has more sugar units, exhibits a poor binding affinity to MelB_{St}.

Structural dynamics measured by HDX-MS

To determine the structural dynamics of MelB_{St} in response to the substrate and/or Na⁺, the bottom-up HDX-MS method was employed to measure the exchange rates of amide hydrogen with deuterium by analyzing time-dependent mass spectra at the peptide level as described (28–30). Three pairs of datasets were collected, including Apo vs. melibiose, Na⁺, or both, as outlined in the Methods section. The analyzed region of MelB_{St} spans positions 2–470 out of 476, achieving a coverage of over 87.47% and 86.62% from 149 or 152 overlapping deuterium-labeled peptides (Table S2; Fig. S4). There are 59, 63, or 59 non-covered residues for the datasets of Apo vs. melibiose-bound, Na⁺-bound, and melibiose- and Na⁺-bound states, respectively. Most of these positions are localized within the transmembrane regions, particularly on the C-terminal half (Table S4). Interestingly, the two longest helices, IV and X, display a significantly higher number of non-covered positions compared to the other helices. While a significant number of such positions are present on the cytoplasmic loops, all periplasmic loops are covered by labeled peptides.

The HDX rates are varying in different regions (Table S4). When averaging the related deuterium uptake (D%) from all peptides at all time points at a given region, the N-terminal helices I and V, as well as the C-terminal helices VIII and IX exhibited higher values, indicating they are more dynamic regions of Apo MelB_{St}.

The differential deuterium labeling (ΔD) was calculated based on the deuterium uptake in the absence (Apo state) or presence of specific ligand(s) (Holo state), and is presented as the residual plot of each peptide between the two states ($D_{\text{Holo}} - D_{\text{Apo}}$) (Fig. 5a) at three labeling time points and the sum of all. The dashed lines indicate the global thresholds (ΔD) for each dataset. The C-terminal His tag and tail residues at positions 470–475 are highly dynamic and fully labeled with greater variations and are excluded from the analysis. The residual plots include data from all peptides, and the ΔD values greater than the global threshold and P values <0.05 are considered statistically significant. Melibiose binding induced a wide range of structural effects on MelB_{St} throughout the entire MelB_{St} polypeptide, and the effects include protection (less deuterium uptake in the Holo state) and deprotection (greater deuterium uptake in the Holo state). In contrast, Na⁺ alone or in combination with melibiose primarily caused deprotection. As shown in Table S3, more than 50% of residues (237, 264, or 257 positions) exhibited either insignificant ΔD values ($D_{\text{Mel}} - D_{\text{Apo}} < |0.186|$, $\Delta D_{(\text{Na}^+)} - D_{\text{Apo}} < |0.224|$, or $\Delta D_{\text{Na}^+(\text{Mel})} - D_{\text{Apo}} < |0.175|$) or greater P values (> 0.05). A peptide with at least one significant ΔD or overlapping peptides extending from the peptides with at least one significant ΔD were mapped onto the melibiose-bound structure of each dataset individually (Fig. S5). The uncovered positions were indicated in gray spheres, and the positions with insignificant ΔD values are shown in backbone representation. Peptides with significant ΔD from all three datasets are presented together in Figure 5b.

Consensus effects are observed from all three conditions. Interestingly, peptides with significant ΔD values are clustered in several regions. Nearly full-length transmembrane helices I and V, as well as the tail helix, demonstrated protections from melibiose, Na⁺, or Na⁺ with melibiose. In particular, the peptides containing sugar-binding critical residues (Lys18, Asp19, and Arg149 on helices I and V) showed statistically high confidence ($P < 0.01$, Fig. 5b, Blue). The peptide 213–226 in the middle loop across the three conditions was also protected. Additionally, the peptide 364–374 on helix XI, which contains sugar-binding residues Gln372 and Thr373, was protected by Na⁺ alone ($P = 0.029$) or Na⁺ with melibiose ($P = 0.011$) and was weakly protected by melibiose but at a P value of 0.056.

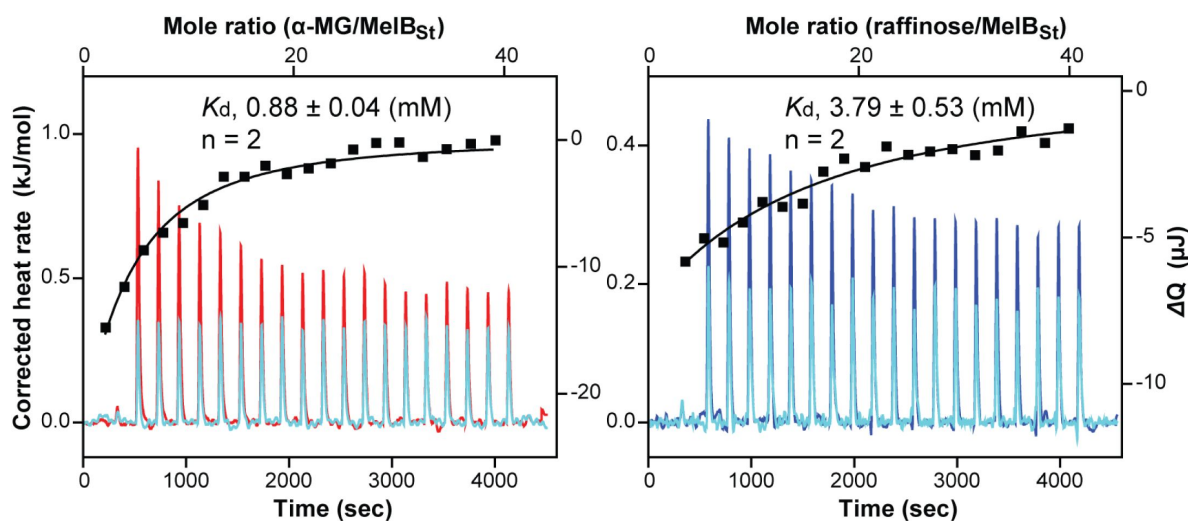


Fig. 4.

Binding affinity determination via ITC.

The binding measurement of α -MG or raffinose to the WT MelB_{St} was performed with a NanoITC calorimeters (TA Instruments) at 25 °C as described in Methods. The thermograms were plotted as baseline-corrected heat rate (μJ /sec; left axis) vs. time (bottom axis) for the titrant to MelB_{St} (red for α -MG and blue for raffinose) or to buffer (light blue). The heat change ΔQ (μJ ; filled black symbol) was plotted against the mole ratio of the sugar to WT MelB_{St} (top/right axes). **(a)** α -MG. **(b)** Raffinose.

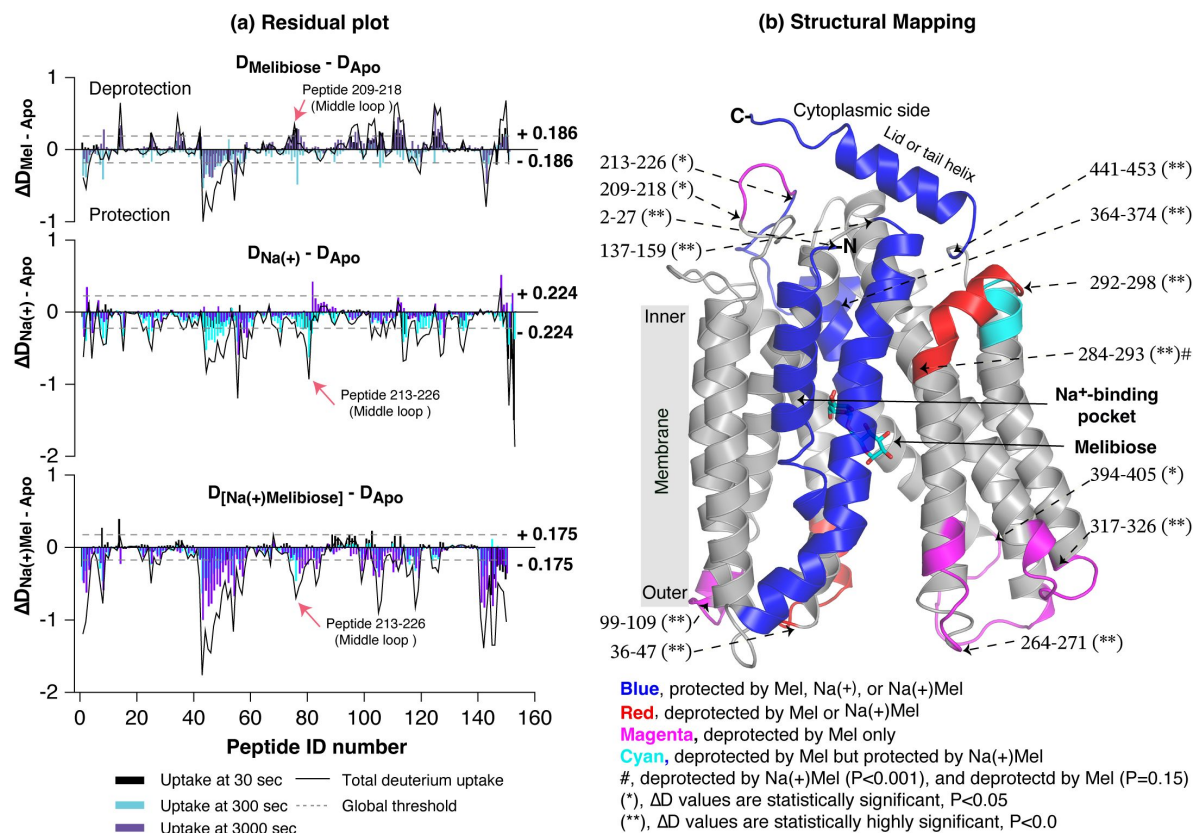


Fig. 5.

HDX-MS.

HDX experiments on WT MelB_{St} in Apo or Holo states (with melibiose, Na⁺, or Na⁺ plus melibiose) were conducted as described in Methods. **(a)** Residual plots ($D_{\text{Holo}} - A_{\text{po}}$). Differential deuterium uptakes ΔD s, calculated from paired conditions for position 2-470, were plotted against overlapping peptides for each time point and the total uptake. Black, cyan, and purple bars, the deuterium uptake at 30, 300, 3000 sec, respectively; dark gray curve, total uptake from all three time points. Deprotection, $\Delta D_{\text{Holo}} - A_{\text{po}} > 0$; protection, $\Delta D_{\text{Holo}} - A_{\text{po}} < 0$. Each sample was analyzed in triplicates. Regions not covered by labeled peptides were excluded. Dashed lines, the global threshold values calculated from each dataset. **(b)** Mapping onto the 3D structure of the melibiose-bound state. Peptides with ΔD value greater than the absolute value of the threshold and $P < 0.05$ at any time point were colored according to the legend on the figure. The membrane region of MelB_{St} is indicated by a gray bar. Each peptide or overlapping peptide was indicated by arrows pointing to the start of the peptide. The helices I, V, VIII, and IX, as well as the C-terminal helix (lid), are labeled individually. Mel, melibiose.

In the presence of melibiose and Na^+ , the deprotections of overlapping peptides 36-47 (loop1-2) and 284-293 (loop8-9) with high confidence ($P < 0.01$) were obtained. Melibiose alone caused the deprotections of peptide 36-47 with medium confidence ($P < 0.05$) (Fig. 5b, red [link](#)). Five deprotected regions induced solely by melibiose are found at the ends of transmembrane helices with extended loops (Fig. 5b, pink [link](#)). Overlapping peptide 292-298 (Fig. 5, Cyan [link](#)) showed opposite effects when comparing melibiose alone to melibiose with Na^+ .

Allosteric effects of Na^+ binding on sugar-binding residues

The helices II and IV, which host the coupling cations, were shown to have significantly lower deuterium uptake (Fig. 5b [link](#); Fig. S5 [link](#)) compared to helices I and V, which are specific for sugar binding and do not participate in cation binding. The effects of deuterium uptake by melibiose, Na^+ , or melibiose plus Na^+ on the residues in the sugar- or Na^+ -binding pockets are summarized in Table S5 [link](#). Seven peptides covering all N-terminal residues for sugar and cation binding were presented with the time courses of deuterium uptake. Notably, peptides 53-62, 120-123, and 121-130, which contain all cation-binding positions (Asp55, Asn58, Asp59, and Thr121), exhibited poor deuterium uptake rates, and their uptake rates were not significantly affected by the binding of melibiose, Na^+ , or both (Fig. 6, labeled in black (II) and blue (IV), respectively [link](#)).

On the contrary, the two peptides (16-20 and 21-27) covering the sugar-binding residues 18, 19, 22, and 26 on helix I exhibited strong protection by melibiose, both in the absence and presence of Na^+ (Fig. 6, magenta label [link](#)). Remarkably, both peptides on helix I are also protected by Na^+ , which binds at helices II and IV, remotely. Additionally, the sugar-binding residue Arg149 on helix V was covered by several overlapping peptides that consistently demonstrated strong protections by melibiose, Na^+ , or melibiose plus Na^+ , as illustrated by peptide 140-151 (Fig. 6, red label [link](#)). The residue 152, located in the binding pocket but not in direct contact with sugar, showed no detectable protection by melibiose or Na^+ alone; however, significant protection was observed in the presence of both melibiose and Na^+ . The protective effects on the C-terminal sugar-binding residues Gln372 and Thr373 have been described; the peptide 369-374 is protected by Na^+ alone and stronger protection was observed in the presence of melibiose plus Na^+ (Fig. 5b [link](#)).

Discussion

The dehydration of sugar molecules is required for sugar binding, yet little is understood about this process in MelB. In the 2.7 Å-resolution α -NPG-bound structure, a water molecule was modeled to a positive density, and surrounded by the OH-4 and OH-6 groups on the galactopyranosyl ring, as well as by Gln372 and Thr373 on helix XI. In the published MelB_{St} structures with α -NPG or DDMB bound (20 [link](#), 22 [link](#)) and the current structures with native melibiose, α -MG, or raffinose bound, due to the limited resolutions, the water molecule was not modeled; however, the positive density at the same position always appeared in the mFo-DFc maps, which indicate that this OH-4 water molecule is part of the sugar binding coordination in MelB.

The structures complexed with galactosides containing different numbers of sugar units, ranging from one to three, clearly demonstrate that the specificity is directed toward the galactosyl moiety (Figs. 1 [link](#)-3 [link](#)). The sugar-protein polar interactions primarily occur at positions 2, 3, 4, and 6. Notably, the orientation of the hydroxyl group at the C4 position is the crucial feature for distinguishing between galactose and glucose. The OH-4 on α -NPG is within hydrogen-bonding distances from the carboxyl group of Asp124 and the indole group of Trp128, and it indirectly connects with Gln372 and Thr373 on helix XI through this water molecule bound to OH-4 (Fig. 2 [link](#)). These interactions define the specificity of galactosyl molecules, as the glucose or glucoside

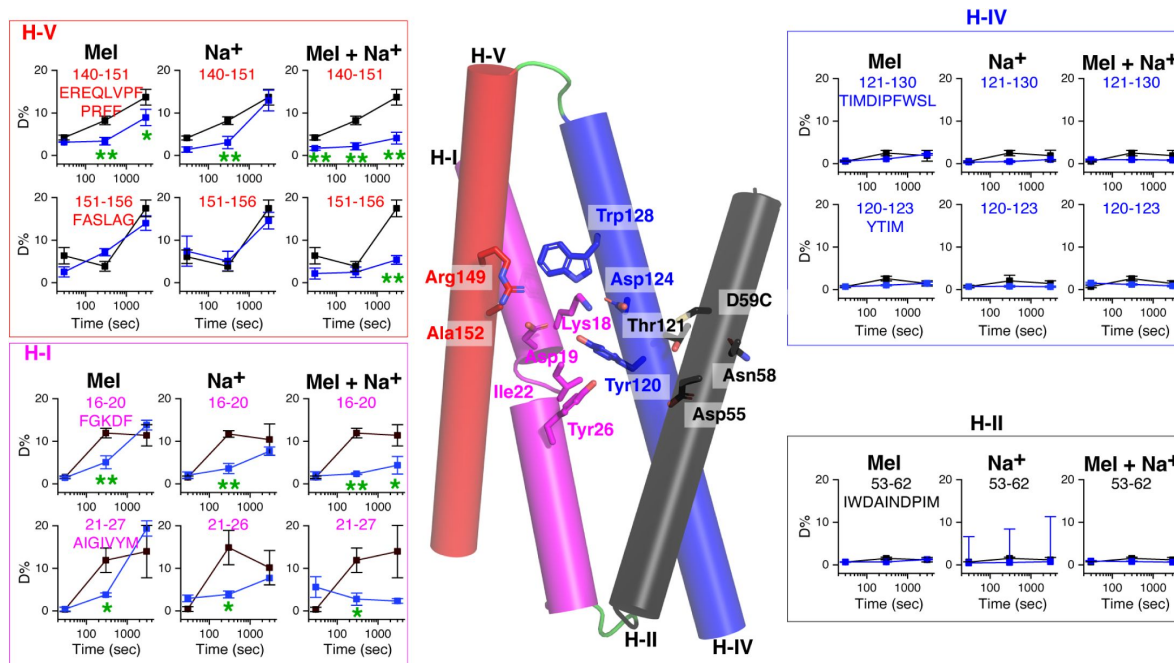


Fig. 6.

Allosteric effects on deuterium uptake by remote binding of Na^+ .

The time course of percentage deuterium uptake was plotted against the labeling time at 0, 30, 300, and 3000 sec for the representative peptides that cover the residues responsible for the specificity of sugar or cation binding in MelB_{ST} . The peptide sequences are shown for each panel. *, statistically significant; **, statistically highly significant. Helices I and V (sugar-binding only), II (cation binding only), and IV (both), were shown in tube representation. Residues involved in sugar binding are represented by sticks and highlighted in magenta (helix I), blue (helix IV), and red (helix V). Residues involved in cation binding are highlighted in black. Labels in the time course and on the model are color-coded, except for the stick representing Thr121, which is black. Helices are labeled with Roman numerals and positioned at the beginning of each helix.

OH-4 points to the opposite side of the galactosyl plane; thus, it does not form the hydrogen-bonding network around OH-4 and therefore would not be selected by MelB_{St}. Clearly, the water molecule bound to OH-4 contributes to the sugar recognition.

Another important piece of structural information is that the OH-4 group and the bound water are in close proximity to the Na⁺-binding residue Thr121 and the cation-binding important residue Lys377, indicating that the OH-4 water molecule appears on the pathway between the two specificity-determining pockets. The OH-3 and OH-2 groups on the opposite edge of the galactopyranosyl ring create multiple hydrogen-bonding interactions with the charged residues Asp19 and Arg149. These interactions likely help to stabilize recognition at OH-4 and enhance the binding affinity.

All available structures of MelB_{St} (19–21) and MD simulations on the coupled translocation of melibiose and Na⁺ (26) indicate that there is no direct interaction between the bound Na⁺ and the bound galactosides. Binding affinity determinations of the primary substrate and the coupling cations demonstrate their positive cooperativity (19, 20, 27). The MD simulations suggested that Na⁺ binding allosterically affects the local electrostatic interactions in the sugar binding pocket. HDX-MS was used to obtain experimental evidence for the structural dynamics induced by melibiose, Na⁺, or melibiose combined with Na⁺.

HDX determines the structural motion or dynamics by measuring exchange rate over time (28). The resolution of this method is limited to the peptide level, and the deuterium uptake rate is affected by the interplay of structural dynamics and solvent accessibility. Previously, we used this technique to study the conformational dynamics between inward- and outward-facing states of MelB_{St} with bound Na⁺. In this study, we tested the Apo state and Holo states with different ligands. At Apo state, the peptides with faster deuterium uptake rates are distributed in both N- and C-terminal domains, and helices I and V are the three helices with the fastest uptake rates, indicating those regions at Apo state are more dynamic. The previous thermal denaturation study detected by circular dichroism spectroscopy showed that the stability of the Apo state is lower than that of any holo states, either with melibiose or Na⁺; the most stable state is the ternary form with both melibiose and Na⁺ bound (27). This HDX study identified the dynamic regions that contribute to the poor stability of Apo MelB_{St}.

There is a significant difference in the dynamics of the sugar-binding helices (I, IV, V, X, and XI) and the cation-binding helices (II and IV). The cation-binding helix II (containing Asp55, Asn58 and Asp59) exhibits poor deuterium uptake and helix IV (containing Thr121) contains more non-covered positions either not covered by labelling or not covered by peptides. In either condition, their solvent accessibility and H/D exchange should be expected to be low. Therefore, the areas hosting the cation binding are relatively rigid (Figs. 5–6). In contrary, most sugar-binding helices, e.g., helices I (containing Lys18, Asp19, Ile22 and Try126), V (Arg149 and Ala152) and XI (containing Gln372 and Thr373), are dynamics; however, helix IV (containing Tyr120, Asp124, and Try128) that also carries the Na⁺-binding residue Thr121 is rigid. Notably, Asp19 and Arg149, which contribute to binding affinity, are located in the most dynamic helices I and V. Stabilization of both helices is expected to increase the affinity for sugar.

Melibiose-induced structural changes include dynamic restraints and enhancements (Fig. S5a). The structural restraints or protections were observed on the binding residue-hosting helices (I and V) and two loop helices (the middle loop and the C-terminal tail), while the deprotections affect a wide range of areas, which are mapped on the helices-loop regions and most are at the periplasmic side. Interestingly, most regions protected by melibiose are also protected by Na⁺ alone; in addition, the water-mediated sugar-binding residues Gln372 and Thr373 are also protected (Fig. S5b). There is no significant deprotection detected by Na⁺, consisting with the conclusion that the Na⁺-bound MelB is more stable than the Apo state.

When both melibiose and Na^+ are presented during the H/D exchange reaction, most of the protected regions by melibiose or Na^+ are also protected (**Fig. S5c**). Two deprotected regions were identified. One is also deprotected by melibiose, which is the peptide 36-47 located in Loop₁₋₂ and the beginning of the helix II; another deprotected region 292-298 is on the extended helix VIII. Both regions become more dynamic in the ternary state with both melibiose and Na^+ . Notably, the deprotected region 292-298 contains Arg295, and two water molecules are modeled in the 2.70 Å map which stabilize the salt-bridge Arg295 and Asp351 on helix X, which is part of the large comprehensive charged network to stabilize the packing between the N- and C-halves at the cytoplasmic leaflet region for the inward-facing state. The binding of both prepared the transporter to initiate the conformational transition from the outward to the inward-facing states. Arg295 might respond to this binding signal and become more dynamics to destabilize the charged network and facilitate the opening at the cytoplasmic surface.

The MD simulation and free energy calculations and distance distribution indicated that the Asp19 is highly dynamic when Na^+ unbound, and the binding of Na^+ shifted the weak interaction of melibiose to stronger single interaction(**26**). HDX-MS data showed that the binding of Na^+ alone causes the protections of the dynamic regions that carry Asp19 and Arg149, so the sugar-binding residues near OH-3 and OH-2 groups are more rigid. Therefore, it can be concluded that Na^+ binding rigidifies the sugar-binding pocket through an allosteric effect on the local electrostatic interactions as proposed from the MD simulations.

The symporter H^+ -coupled lactose permease LacY and MelB share several common features (**5**, **31**, **32**). Both transporters share a similar substrate spectrum, and all are selective for galactosides. In addition, the cation-binding affinity in both LacY and MelB does not appear to be influenced by conformational transition(**21**, **32**, **33**). Furthermore, their cation sites do not have direct contact with their primary substrate galactosides. Glu325 of LacY is the sole protonation residue (**34**) and located in a hydrophobic milieu with no direct contact with galactosides(**31**, **33**, **35**). It has also been demonstrated that the sugar-binding affinity depends on the protonation status of LacY (**36**), which could be postulated to be an allosteric effect. The role of the coupling cation as an allosteric activator in cation-coupled symporters might be a common mechanism. Understanding this could provide broader insights into how these transport systems operate.

Experimental Procedures

Reagents

Melibiose was purchased from *Acros Organics (ThermoFisher Scientific)*. Raffinose (RPI Chemicals), α -methyl galactoside and *p*-nitrophenyl galactoside are purchased from Sigma-Aldrich. Detergents undecyl- β -D-maltopyranoside (UDM) and DDM (dodecyl- β -D-maltopyranoside (DDM) were purchased from *Anatrace*. *E. coli* lipids (Extract Polar, 100600) were purchased from *Avanti Polar Lipids, Inc.* All other materials were reagent grade and obtained from commercial sources.

Strains and plasmids

E. coli DW2 cells (*melA*⁺, *melB*⁺, *lacZ*^Y) (**37**) were used for protein expression and functional studies. The expression plasmids pK95/ Δ AH/WT MelB_{St}/CHis₁₀(**15**) and pK95/ Δ AH/D59CMelB_{St}/CHis₁₀ (**18**) were used for constitutive expression.

MelB_{St} protein expression and purification

Cell growth for the large-scale production of WT MelB_{St} or D59C MelB_{St} were carried out in *E. coli* DW2 cells (**18**, **38**). Briefly, MelB_{St} purification by cobalt-affinity chromatography (Talon Superflow Metal Affinity Resin, Takara) after extraction by 1.5% UDM. MelB_{St} protein was eluted

with 250 mM imidazole in a buffer containing 50 mM NaPi, pH 7.5, 200 mM NaCl, 0.035% UDM, and 10% glycerol, and further dialyzed to change the buffer conditions accordingly.

Protein concentration assay

The Micro BCA Protein Assay (Pierce Biotechnology, Inc.) was used to determine the protein concentration.

Isothermal titration calorimetry

All ITC ligand-binding assays were performed with the TA Instruments (Nano-ITC device) as described (19), which yields the exothermic binding as a positive peak. The MelB_{St} was dialyzed overnight with assay buffer containing 20 mM Tris-HCl (pH7.5), 100 mM NaCl, 10% glycerol and 0.035% UDM. The ligands are prepared by dissolving in the same batch of dialysis buffer for buffer matching. In a typical experiment, the titrand (MelB_{St}) placed in the ITC Sample Cell was titrated with the specified titrant raffinose, α -methyl galactoside, melibiose, or α -NPG (placed in the Syringe) in the assay buffer by an incremental injection of 2.5- μ L aliquots at an interval of 2500 sec at a constant stirring rate of 250 rpm (nano-ITC). MelB_{St} protein samples were buffer-matched to the assay buffer by dialysis. The normalized heat changes were subtracted from the heat of dilution elicited by the last few injections, where no further binding occurred, and the corrected heat changes were plotted against the mole ratio of titrant versus titrand. The values for the binding association constant (K_a) were obtained by fitting the data using the one-site independent-binding model included in the NanoAnalyze software (version 3.7.5). The dissociation constant (K_d) = $1/K_a$.

Crystallization, native diffraction data collection, and processing

The D59C MelB_{St} dialyzed overnight against the sugar-free dialysis buffer (20 mM Tris-HCl, pH 7.5, 100 mM NaCl, 0.035% UDM, and 10% glycerol), concentrated with Vivaspinn column at 50 kDa cutoff, and stored at -80°C . A phospholipid stock solution of 20 mM was prepared by dissolving the *E. coli* Extract Polar (Avanti, 100600) with a dialysis buffer containing 0.01% DDM. The protein sample was diluted to a final concentration of 10 mg/ml with the same sugar-free dialysis buffer, supplemented with phospholipids at 3.6 mM and 30 mM of melibiose or α -MG, 40 mM raffinose, or 6 mM of α -NPG in DMSO solution. Crystallization trials were carried out by the hanging-drop vapor-diffusion method at 23°C by mixing 2- μ L protein with 2- μ L reservoir.

Crystals from D59C MelB_{St} protein with the melibiose, α -MG, or α -NPG appeared against a reservoir consisting of 100 mM Tris-HCl, pH 8.5, 100 mM NaCl₂, 50 mM CaCl₂, and 32-35% PEG 400. For the raffinose-containing sample, the crystals were collected from 100 mM Tris-HCl, pH 8.5, 50 mM CaCl₂, 50 mM BaCl₂, and 32.5% PEG 400. All crystals were frozen with liquid nitrogen in 2 weeks, and tested for X-ray diffraction at the Lawrence Berkeley National Laboratory ALS BL 5.0.1 (dataset for melibiose, α -MG, and α -NPG-containing complex) or 5.0.2 (raffinose-containing datasets) via remote data collection method.

ALS auto-processing XDS or DIALS programs output files were further reduction by AIMLESS in the ccp4i2 program for the structure solution (39). The statistics in data collection are described in Table S1.

Structure determination

The structure determination was performed by the Molecular Replacement method using the α -NPG-bound D59C MelB_{St} mutant structure [PDB ID 7L17] as the search template, followed by rounds of manual building and refinement to resolutions of 2.70 Å, 3.05, 3.45, or 3.68 Å for structure with α -NPG, melibiose, raffinose, or α -MG, respectively, in Phenix (40). The structures were modeled from positions 2 to 453 or 456, respectively, without a gap, and the missing side chains due to the density disorder, and Ramachandran assessment were listed in the Tables S6.

Sugar docking and modeling

A strong positive density, with varying sizes and shapes, was observed in the difference maps of each of the four structures. The size and shape matched the sugars that co-crystallized, and the docked sugar molecules fitted well with the densities. The sugar refinement restraints were generated from SMILES using the ELBOW program in Phenix. To the 2.7 Å α -NPG-bound map, six water molecules were modeled. In addition, a PEG molecule (ligand ID 1PE) was also model to a strong positive density with a sausage shape aligning with the helix IX.

Hydrogen-deuterium exchange coupled to mass spectrometry (HDX-MS)

In-solution HDX-MS experiment was performed to study the substrate-induced structural dynamics of MelB_{St}. As described (21 [↗](#)), the labeling, quenching, lipids removal, and online digestion were achieved using a fully automated manner using a HDx3 extended parallel system (LEAP Technologies, Morrisville, NC)(41 [↗](#), 42 [↗](#)). MelB_{St} was prepared at 50.0 μ M in a Na⁺-free buffer (25 mM Tris-HCl, pH 7.5, 150 mM choline chloride, 10% glycerol, 0.035% UDM in H₂O), either in the absence of a substrate (Apo) or in the presence of 100 mM melibiose, 100 mM NaCl, or both. The hydrogen/deuterium exchange reaction as described previously (21 [↗](#)). Briefly, aliquots of 4 μ l of each sample were diluted 10-fold into the labeling buffer (25 mM Tris-HCl, pH 7.5, 50 mM choline chloride, 10% glycerol, 0.035% UDM in D₂O), without or with 100 mM of melibiose, Na⁺, or both. Labeled samples were incubated in D₂O buffer at 20 °C for multiple time points (30 sec, 300 sec, and 3000 sec) in triplicates and non-deuterated controls were prepared in a similar manner except H₂O buffer was used in the labeling step.

At each designated time point, reaction was quenched by adding an equal volume of ice-cold quench buffer (6 M urea, 100 mM citric acid, pH, 2.3 in H₂O) for 180 seconds at 0 °C and immediately subjected to a lipid filtration module integrated on the LEAP PAL system. After incubation of a 60 sec with ZrO₂ particles, the LEAP X-Press then compressed the filter assembly to separate proteins from the ZrO₂ particles-bound phospholipids and detergents. The filtered protein sample was injected into a cool box for online digestion and separation.

LC/MS bottom-up HDX was performed using a Thermo Scientific™ Ultimate™ 3000 UHPLC system and Thermo Scientific™ Orbitrap Eclipse™ Tribrid™ mass spectrometer. Samples were digested with a Nepenthesin-2 (Affipro, Czech Republic) column at 8 °C and then trapped in a 1.0 mm x 5.0 mm, 5.0 μ m trap cartridge for desalting over 180 sec. The resulting peptides were then separated on a Thermo Scientific™ Hypersil Gold™, 50 x1 mm, 1.9 μ m, C18 column with a gradient of 10 % to 40 % gradient (A: water, 0.1 % formic acid; B: acetonitrile, 0.1 % formic acid) for 15 minutes at a flow rate of 40 μ L/min. A pepsin wash was added in between runs to minimize the carryover.

A nonspecific digested peptide database has been created for MelB_{St} with a separate MS/MS measurement of non-deuterated samples as described (21 [↗](#)). Digested peptides from undeuterated MelB_{St} protein were identified on the orbitrap mass spectrometer using the same LC gradient as the HDX-MS experiment. Using the Thermo BioPharma Finder software (v 5.1), MS2 spectra were matched to the MelB_{St} sequence with fixed modifications.

A number of 146 or 150 peptide assignments (confident HDX data cross all labeling times) were confirmed for MelB_{St} samples giving 86-87% sequence coverage. The MS data were processed using the Sierra Analytics HDEaminer software with the MelB_{St} peptide database. Following the automated HDX-MS analysis, manual curation was performed. Upon the completion of the data review, a single charge state with high-quality spectra for all replicates across all HDX labeling times was chosen to represent HDX for each peptide. Differential HDX data were tested for statistical significance using the hybrid significance testing criteria method with an in-house MATLAB script, where the HDX differences at different protein states were calculated ($\Delta D = D_{\text{Holo}} -$

D_{Apo}). Mean HDX differences from the three replicates were assigned as significant according to the hybrid criteria based on the pooled standard deviation and Welch's t-test with $P < 0.05$. The statistically significant differences observed at each residue ($|D_{\text{Mel-Apo}} - D_{\text{Apo}}| < 0.186$, $|\Delta D_{(\text{Na}^+)} - D_{\text{Apo}}| < 0.224$, or $|\Delta D_{\text{Na}^+(\text{Mel})} - D_{\text{Apo}}| < 0.175$) were used to map HDX consensus effects based on overlapping peptides onto the structure models.

Statistics and reproducibility

All experiments were performed 2–4 times. The average values were presented in the table with standard errors. An unpaired t-test was used for statistical analysis.

Graphs

Pymol (3.1.5.1) ([43](#)) and UCSF ChimeraX (X) were used to generate all graphs. The program Origin 2024 was used to plot the ITC curves.

Data availability statement

The x-ray diffraction datasets and models have been deposited to wwPDB under the accession codes 9OLD for the α -nitrophenyl galactoside-bound, 9OLI for the melibiose-bound, 9OLR for the α -methyl galactoside, as well as 9OLP for the raffinose-bound D59C MelB_{St}.

Acknowledgements

The x-ray diffraction datasets were collected at ALS BL 5.01 or 5.0.2. This work was supported by the National Institutes of Health Grant R35GM153222 to L.G.

Additional information

- MelB_{St}: *Salmonella enterica* serovar Typhimurium melibiose permease
- MFS: major facilitator superfamily
- ITC: isothermal titration calorimetry
- α -NPG: 4-nitrophenyl- α -D-galactopyranoside
- Mel: melibiose
- α -MG: α -methyl galactoside
- Raff: raffinose
- K_d : dissociation constant
- HDX-MS: hydrogen-deuterium exchange coupled to mass spectrometry
- D: deuterium

Funding

National Institutes of Health (R35GM153222)

Additional files

Supplemental Tables 1–6 [↗](#)

Supplemental Figures 1-5 [↗](#)

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Reviewer #1 (Public review):

While the structure of the melibiose permease in both outward and inward-facing forms has been solved previously, there remain unanswered questions regarding its mechanism. Hariharan et al set out to address this with further crystallographic studies complemented with ITC and hydrogen-deuterium exchange (HDX) mass spectrometry. They first report 4 different crystal structures of galactose derivatives to explore molecular recognition, showing that the galactose moiety itself is the main source of specificity. Interestingly, they observe a water-mediated hydrogen bonding interaction with the protein and suggest that this water molecule may be important in binding.

The results from the crystallography appear sensible, though the resolution of the data is low, with only the structure with NPG better than 3Å. However, it is a bit difficult to understand what novel information is being brought out here and what is known about the ligands. For instance, are these molecules transported by the protein or do they just bind? They measure the affinity by ITC, but draw very few conclusions about how the affinity correlates with the binding modes. Can the protein transport the trisaccharide raffinose?

The HDX also appears to be well done; however, in the manuscript as written, it is difficult to understand how this relates to the overall mechanism of the protein and the conformational changes that the protein undergoes.

<https://doi.org/10.7554/eLife.108335.1.sa3>

Reviewer #2 (Public review):

This manuscript from Hariharan, Shi, Viner, and Guan presents x-ray crystallographic structures of membrane protein MelB and HDX-MS analysis of ligand-induced dynamics. This work improves on the resolution of previously published structures, introduces further sugar-bound structures, and utilises HDX to explore in further depth the previously observed positive cooperatively to cotransported cation Na⁺. The work presented here builds on years of previous study and adds substantial new details into how Na⁺ binding facilitates melibiose binding and deepens the fundamental understanding of the molecular basis underlying the symport mechanism of cation-coupled transporters. However, the presentation of the data lacks clarity, and in particular, the HDX-MS data interpretation requires further explanation in both methodology and discussion.

Comments on Crystallography and biochemical work:

(1) It is not clear what Figure 2 is comparing. The text suggests this figure is a comparison of the lower resolution structure to the structure presented in this work; however, the figure legend does not mention which is which, and both images include a modelled water molecule that was not assigned due to poor resolution previously, as stated by the authors, in the previously generated structure. This figure should be more clearly explained.

(2) It is slightly unclear what the ITC measurements add to this current manuscript. The authors comment that raffinose exhibiting poor binding affinity despite having more sugar units is surprising, but it is not surprising to me. No additional interactions can be mapped to these units on their structure, and while it fits into the substrate binding cavity, the extra bulk of additional sugar units is likely to reduce affinity. In fact, from their listed ITC measurements, this appears to be the trend. Additionally, the D59C mutant utilised here in structural determination is deficient in sodium/cation binding. The reported allostery of sodium-sugar binding will likely influence the sugar binding motif as represented by these structures. This is clearly represented by the authors' own ITC work. The ITC included in this work was carried out on the WT protein in the presence of Na⁺. The authors could benefit from clarifying how this work fits with the structural work or carrying out ITC with the D59C mutant, or additionally, in the absence of sodium.

Comments on HDX-MS work:

While the use of HDX-MS to deepen the understanding of ligand allostery is an elegant use of the technique, this reviewer advises the authors to refer to the Masson et al. (2019) recommendations for the HDX-MS article (<https://doi.org/10.1038/s41592-019-0459-y>) on how to best present this data. For example:

(1) The Methodology includes a lipid removal step. Based on other included methods, I assumed that the HDX-MS was being carried out in detergent-solubilised protein samples. I therefore do not see the need for a lipid removal step that is usually included for bilayer reconstituted samples. I note that this methodology is the same as previously used for MelB. It should be clarified why this step was included, if it was in fact used, aka, further details on the sample preparation should be included.

(2) A summary of HDX conditions and results should be given as recommended, including the mean peptide length and average redundancy per state alongside other included information such as reaction temperature, sequence coverage, etc., as prepared for previous publications from the authors, i.e., Hariharan et al., 2024.

(3) Uptake plots per peptide for the HDX-MS data should be included as supporting information outside of the few examples given in Figure 6.

(4) A reference should be given to the hybrid significance testing method utilised. Additionally, as stated by Hageman and Weis (2019) (doi:10.1021/acs.analchem.9b01325), the use of $P < 0.05$ greatly increases the likelihood of false positive ΔD identifications. While the authors include multiple levels of significance, what they refer to as high and lower significant results, this reviewer understands that working with dynamic transporters can lead to increased data variation; a statement of why certain statistical criteria were chosen should be included, and possibly accompanied by volcano plots. The legend of Figure 6 should include what P value is meant by * and ** rather than statistically significant and highly statistically significant.

(5) Line 316 states a significant difference in seen in dynamics, how is significance measured here? There is no S.D. given in Table S4. Can the authors further comment on the potential involvement in solvent accessibility and buried helices that might influence the overall dynamics outside of their role in sugar vs sodium binding? An expected low rate of exchange suggests that dynamics are likely influenced by solvent accessibility or peptide hydrophobicity? The increased dynamics at peptides covering the Na binding site on overall more dynamic helices suggests that there is no difference between the dynamics of each site.

(6) Previously stated HDX-MS results of MelB (Hariharan et al., 2024) state that the transmembrane helices are less dynamic than polypeptide termini and loops with similar distributions across all transmembrane bundles. The previous data was obtained in the presence of sodium. Does this remove the difference in dynamics in the sugar-binding helices and the cation-binding helices? Including this comparison would support the statement that the sodium-bound MelB is more stable than the Apo state, along with the lack of deprotection observed in the differential analysis.

(7) Have the authors considered carrying out an HDX-MS comparison between the WT and the D59C mutant? This may provide some further information on the WT structure (particularly a comparison with sugar-bound). This could be tied into a nice discussion of their structural data.

(8) Have the authors considered utilising Li^+ to infer how cation selectivity impacts the allostery? Do they expect similar stabilisation of a higher-affinity sugar binding state with all cations?

(9) MD of MelB suggests all transmembrane helices are reorientated during substrate translocation, yet substrate and cotransporter ligand binding only significantly impacts a small number of helices. Can the authors comment on the ensemble of states expected from each HDX experiment? The data presented here instead shows overall stabilisation of the transporter. This data can be compared to that of HDX on MFS sugar cation symporter Xyle, where substrate binding induces a transition to OF state. There is no discussion of how this HDX data compares to previous MFS sugar transporter HDX. The manuscript could benefit from this comparison rather than a comparison to LacY. It is unlikely that there are universal mechanisms that can be inferred even from these model proteins. Highlighting differences instead between these transport systems provides broader insights into this protein class. Doi: 10.1021/jacs.2c06148 and 10.1038/s41467-018-06704-1.

(10) Additionally, the recent publication of SMFS data (by the authors: doi:10.1016/j.str.2022.11.011) states the following: "In the presence of either melibiose or a coupling Na⁺-cation, however, MelB increasingly populates the mechanically less stable state which shows a destabilized middle-loop C3." And "In the presence of both substrate and co-substrate, this mechanically less stable state of MelB is predominant.". It would benefit the authors to comment on these data in contrast to the HDX obtained here. Additionally, is the C3 loop covered, and does it show the destabilization suggested by these studies? HDX can provide a plethora of results that are missing from the current analysis on ligand allostery. The authors instead chose to reference CD and thermal denaturation methods as comparisons.

<https://doi.org/10.7554/eLife.108335.1.sa2>

Reviewer #3 (Public review):

Summary:

The melibiose permease from *Salmonella enterica* serovar Typhimurium (MelBSt) is a member of the Major Facilitator Superfamily (MFS). It catalyzes the symport of a galactopyranoside with Na⁺, H⁺, or Li⁺, and serves as a prototype model system for investigating cation-coupled transport mechanisms. In cation-coupled symporters, a coupling cation typically moves down its electrochemical gradient to drive the uphill transport of a primary substrate; however, the precise role and molecular contribution of the cation in substrate binding and translocation remain unclear. In a prior study, the authors showed that the binding affinity for melibiose is increased in the presence of Na⁺ by about 8-fold, but the molecular basis for the cooperative mechanism remains unclear. The objective of this study was to better understand the allosteric coupling between the Na⁺ and melibiose binding sites. To verify the sugar-recognition specific determinants, the authors solved the outward-facing crystal structures of a uniport mutant D59C with four sugar ligands containing different numbers of monosaccharide units (α -NPG, melibiose, raffinose, or α -MG). The structure with α -NPG bound has improved resolution (2.7 Å) compared to a previously published structure and to those with other sugars. These structures show that the specificity is clearly directed toward the galactosyl moiety. However, the increased affinity for α -NPG involves its hydrophobic phenyl group, positioned at 4 Å-distance from the phenyl group of Tyr26 forms a strong stacking interaction. Moreover, a water molecule bound to OH-4 in the structure with α -NPG was proposed to contribute to the sugar recognition and appears on the pathway between the two specificity-determining pockets. Next, the authors analyzed by hydrogen-to-deuterium exchange coupled to mass spectrometry (HDX-MS) the changes in structural dynamics of the transporter induced by melibiose, Na⁺, or both. The data support the conclusion that the binding of the coupling cation at a remote location stabilizes the sugar-binding residues to switch to a higher-affinity state. Therefore, the coupling cation in this symporter was proposed to be an allosteric activator.

Strengths:

- (1) The manuscript is generally well written.
- (2) This study builds on the authors' accumulated knowledge of the melibiose permease and integrates structural and HDX-MS analyses to better understand the communication between the sodium ion and sugar binding sites. A high sequence coverage was obtained for the HDX-MS data (86-87%), which is high for a membrane protein.

Weaknesses:

(1) I am not sure that the resolution of the structure (2.7 Å) is sufficiently high to unambiguously establish the presence of a water molecule bound to OH-4 of the α-NPG sugar. In Figure 2, the density for water 1 is not obvious to me, although it is indeed plausible that water mediates the interaction between OH4/OH6 and the residues Q372 and T373.

(2) Site-directed mutagenesis could help strengthen the conclusions of the authors. Would the mutation(s) of Q372 and/or T373 support the water hypothesis by decreasing the affinity for sugars? Mutations of Thr 121, Arg 295, combined with functional and/or HDX-MS analyses, may also help support some of the claims of the authors regarding the allosteric communication between the two substrate-binding sites.

(3) The main conclusion of the authors is that the binding of the coupling cation stabilizes those dynamic sidechains in the sugar-binding pocket, leading to a high-affinity state. This is visible when comparing panels c and a from Figure S5. However, there is both increased protection (blue, near the sugar) and decreased protection in other areas (red). The latter was less commented, could the increased flexibility in these red regions facilitate the transition between inward- and outward-facing conformations? The HDX changes induced by the different ligands were compared to the apo form (see Figure S5). It might be worth it for data presentation to also analyze the deuterium uptake difference by comparing the conditions sodium ion+melibiose vs melibiose alone. It would make the effect of Na⁺ on the structural dynamics of the melibiose-bound transporter more visible. Similarly, the deuterium uptake difference between sodium ion+melibiose vs sodium ion alone could be analyzed too, in order to plot the effect of melibiose on the Na⁺-bound transporter.

(4) For non-specialists, it would be beneficial to better introduce and explain the choice of using D59C for the structural analyses.

(5) In Figure 5a, deuterium changes are plotted as a function of peptide ID number. It is hardly informative without making it clearer which regions it corresponds to. Only one peptide is indicated (213-226), I would recommend indicating more of them in areas where deuterium changes are substantial.

(6) From prior work of the authors, melibiose binding also substantially increases the affinity of the sodium ion. Can the authors interpret this observation based on the HDX data?

<https://doi.org/10.7554/eLife.108335.1.sa1>

Author response:

Reviewer #1:

While the structure of the melibiose permease in both outward and inward-facing forms has been solved previously, there remain unanswered questions regarding its mechanism. Hariharan et al set out to address this with further crystallographic studies complemented with ITC and hydrogen-deuterium exchange (HDX) mass spectrometry.

They first report 4 different crystal structures of galactose derivatives to explore molecular recognition, showing that the galactose moiety itself is the main source of specificity. Interestingly, they observe a water-mediated hydrogen bonding interaction with the protein and suggest that this water molecule may be important in binding.

We appreciate the understanding of our work presented in this manuscript by this reviewer.

The results from the crystallography appear sensible, though the resolution of the data is low, with only the structure with NPG better than 3Å. However, it is a bit difficult to

understand what novel information is being brought out here and what is known about the ligands. For instance, are these molecules transported by the protein or do they just bind? They measure the affinity by ITC, but draw very few conclusions about how the affinity correlates with the binding modes. Can the protein transport the trisaccharide raffinose?

The four structures with a bound sugar of different sizes aimed to identify the binding motif on both the primary substrate (sugar) and the transporter (MelB_{St}). Although the resolutions of the structures complexed with melibiose, raffinose, or α-MG are relatively low, the size and shape of the densities at each structure are consistent with the corresponding sugar molecules, which provide valuable data for determining the pose of the bound sugar. Additionally, there is another α-NPG-bound structure at a higher resolution of 2.7 Å. Therefore, our new data support the published binding site with the galactosyl moiety as the main interacting group. The identified water-1 in this study further confirms the orientation of C4-OH. Notably, this transporter does not recognize or transport glucosides where the orientation of C4-OH at the glucopyranosyl ring is opposite. We will provide stronger data to support the water-1.

Regarding the raffinose question, we should have clearly introduced the historical background. Bacterial disaccharide transporters have broad specificity, allowing them to work on a group of sugars with shared structural elements; for example, one sugar molecule can be transported by several transporters. As reported in the literature, the galactosides melibiose, lactose, and raffinose can be transported by both LacY and MelB of *E. coli*. We did not test whether MelB_{St} can transport the α-NPG and raffinose. To address this issue and strengthen our conclusions, we plan to conduct additional experiments to gather evidence of the translocation of these sugars by MelB_{St}.

The HDX also appears to be well done; however, in the manuscript as written, it is difficult to understand how this relates to the overall mechanism of the protein and the conformational changes that the protein undergoes.

Previously, we used HDX-MS to examine the conformational transition between inward- and outward-facing conformations using a conformation-specific nanobody to trap MelB_{St} in an inward-facing state, as structurally resolved by cryoEM single-particle analysis and published in eLife 2024. That study identified dynamic regions that may be involved in the conformational transitions; however, there was no sugar present. We also solved and published the crystal structure of the apo D59C MelB_{St}. The sugar-bound and apo states are virtually identical. To address the positive cooperativity of binding between the sugar and co-transport cations observed in biophysical analysis, in this study, we utilize HDX-MS to analyze the structural dynamics induced by melibiose, Na⁺, or both, focusing on the binding residues at the sugar-binding and cation-binding pockets. The results suggest that the coupling cation stabilizes sugar-binding residues at helices I and V, contributing to affinity but not specificity.

Since MelB_{St} favors the outward-facing conformation, and simulations on the free-energy landscape suggest that the highest affinity of the sugar-bound state is also at an outward-facing state, MelB_{St} in both the apo and bound states tend to remain in the outward-facing conformation. We will include a section comparing these differences. Thank you to this reviewer for the critical insight.

Reviewer #2:

This manuscript from Hariharan, Shi, Viner, and Guan present x-ray crystallographic structures of membrane protein MelB and HDX-MS analysis of ligand-induced dynamics. This work improves on the resolution of previously published structures, introduces further sugar-bound structures, and utilises HDX to explore in further depth the

previously observed positive cooperatively to cotransported cation Na^+ . The work presented here builds on years of previous study and adds substantial new details into how Na^+ binding facilitates melibiose binding and deepens the fundamental understanding of the molecular basis underlying the symport mechanism of cation-coupled transporters. However, the presentation of the data lacks clarity, and in particular, the HDX-MS data interpretation requires further explanation in both methodology and discussion.

We thank this reviewer for taking the time to read our previous articles related to this manuscript.

Comments on Crystallography and biochemical work:

(1) It is not clear what Figure 2 is comparing. The text suggests this figure is a comparison of the lower resolution structure to the structure presented in this work; however, the figure legend does not mention which is which, and both images include a modelled water molecule that was not assigned due to poor resolution previously, as stated by the authors, in the previously generated structure. This figure should be more clearly explained.

This figure shows a stereo view of a density map created in cross-eye style to demonstrate its quality. We will update this figure with a higher-resolution map, and the density for Wat-1 is clearly visible. This also addresses Reviewer-3's comment regarding the map resolution.

(2) It is slightly unclear what the ITC measurements add to this current manuscript. The authors comment that raffinose exhibiting poor binding affinity despite having more sugar units is surprising, but it is not surprising to me. No additional interactions can be mapped to these units on their structure, and while it fits into the substrate binding cavity, the extra bulk of additional sugar units is likely to reduce affinity. In fact, from their listed ITC measurements, this appears to be the trend. Additionally, the D59C mutant utilised here in structural determination is deficient in sodium/cation binding. The reported allostery of sodium-sugar binding will likely influence the sugar binding motif as represented by these structures. This is clearly represented by the authors' own ITC work. The ITC included in this work was carried out on the WT protein in the presence of Na^+ . The authors could benefit from clarifying how this work fits with the structural work or carrying out ITC with the D59C mutant, or additionally, in the absence of sodium.

While raffinose and α -MG have been reported as substrates of MelB in *E. coli*, binding data are unavailable; additionally, for MelB_{ST}, we lack data on the binding of two of the four sugars or sugar analogs. We performed a label-free binding assay using ITC to address this concern with the WT MelB_{ST}. We will also perform the binding assay with the D59C MelB_{ST}, since sugar binding has been structurally analyzed with this mutant, as pointed out by this reviewer. Along with other new functional results, we will prepare a new Figure 1 on functional analysis, which will also address the comment regarding extra bulk at the non-galactosyl moiety with poor affinity.

This D59C uniport mutant exhibits increased thermostability, making it a valuable tool for crystal structure determination, especially since the wild type (WT) is difficult to crystallize at high quality. Asp59 is the only site that responds to the binding of all coupling cations: Na^+ , Li^+ , or H^+ . Notably, this mutant selectively abolishes cation binding and cotransport. However, it still maintains intact sugar binding with slightly higher affinity and preserves the conformational transition, as demonstrated by an electroneutral transport reaction, the melibiose exchange, and fermentation assays with intact cells. Therefore, the structural data derived from this mutant are significant and offer important mechanistic insights into sugar transport. We will provide additional details during the revision.

Comments on HDX-MS work:

While the use of HDX-MS to deepen the understanding of ligand allostery is an elegant use of the technique, this reviewer advises the authors to refer to the Masson et al. (2019) recommendations for the HDX-MS article (<https://doi.org/10.1038/s41592-019-0459-y>) on how to best present this data. For example:

All authors appreciate this reviewer's comments and suggestions, which will be incorporated into the revision.

(1) The Methodology includes a lipid removal step. Based on other included methods, I assumed that the HDX-MS was being carried out in detergent-solubilised protein samples. I therefore do not see the need for a lipid removal step that is usually included for bilayer reconstituted samples. I note that this methodology is the same as previously used for MelB. It should be clarified why this step was included, if it was in fact used, aka, further details on the sample preparation should be included.

Yes, a lipid/detergent removal step was applied in this study and in previous studies and this information was clearly described in Methods.

(2) A summary of HDX conditions and results should be given as recommended, including the mean peptide length and average redundancy per state alongside other included information such as reaction temperature, sequence coverage, etc., as prepared for previous publications from the authors, i.e., Hariharan et al., 2024.

We will update the Table S2. Thank you.

(3) Uptake plots per peptide for the HDX-MS data should be included as supporting information outside of the few examples given in Figure 6.

We will prepare the plots in supplementary information.

*(4) A reference should be given to the hybrid significance testing method utilised. Additionally, as stated by Hageman and Weis (2019) (doi:10.1021/acs.analchem.9b01325), the use of $P < 0.05$ greatly increases the likelihood of false positive ΔD identifications. While the authors include multiple levels of significance, what they refer to as high and lower significant results, this reviewer understands that working with dynamic transporters can lead to increased data variation; a statement of why certain statistical criteria were chosen should be included, and possibly accompanied by volcano plots. The legend of Figure 6 should include what P value is meant by * and ** rather than statistically significant and highly statistically significant.*

We appreciate this comment and will cite this article on the hybrid significance method. We will include volcano plots for each dataset. We fully acknowledge that using a cutoff of $P < 0.05$ can increase the likelihood of false-positive identifications. However, given the complexity of the samples analyzed in this study, we believe that some important changes may have been excluded due to higher variability within the dataset. By applying multiple levels of statistical testing, we determined that $P < 0.05$ represents a suitable threshold for this study. The threshold values were marked in the residual plots and explained in the text. For Figure 6, we have revised it by showing the P value directly.

(5) Line 316 states a significant difference in seen in dynamics, how is significance measured here? There is no S.D. given in Table S4. Can the authors further comment on the potential involvement in solvent accessibility and buried helices that might influence the overall dynamics outside of their role in sugar vs sodium binding? An expected low rate of exchange suggests that dynamics are likely influenced by solvent accessibility or peptide hydrophobicity? The increased dynamics at peptides covering the Na binding site on overall more dynamic helices suggests that there is no difference between the dynamics of each site.

Table S4 was created to provide an overall view of the dynamic regions. If we understand correctly, this reviewer asked us to comment on the effect of solvent accessibility or hydrophobic regions on the overall dynamics outside the binding residues of the peptides that carry binding residues. Since the HDX rate is influenced by two linked factors: solvent accessibility and hydrogen-bonding interactions that reflect structural dynamics, poor solvent accessibility in buried regions results in low deuterium uptakes. The peptides in our dataset that include the Na⁺-binding site showed low HDX, likely due to poor solvent accessibility and structural stability. It is unclear what this reviewer meant by "increased dynamics at peptides covering the Na binding site on overall more dynamic helices." We do not observe increased dynamics in peptides covering Na⁺-binding sites.

(6) Previously stated HDX-MS results of MelB (Hariharan et al., 2024) state that the transmembrane helices are less dynamic than polypeptide termini and loops with similar distributions across all transmembrane bundles. The previous data was obtained in the presence of sodium. Does this remove the difference in dynamics in the sugar-binding helices and the cation-binding helices? Including this comparison would support the statement that the sodium-bound MelB is more stable than the Apo state, along with the lack of deprotection observed in the differential analysis.

Thanks for this suggestion. The previous datasets were collected in the presence of Na⁺. In the current study, we also have a Na-containing dataset. Both showed similar results: the multiple overlapping peptides covering the sugar-binding residues on helices I and V have higher HDX rates than those covering the Na⁺-binding residues, even when Na⁺ is present in both datasets.

(7) Have the authors considered carrying out an HDX-MS comparison between the WT and the D59C mutant? This may provide some further information on the WT structure (particularly a comparison with sugar-bound). This could be tied into a nice discussion of their structural data.

Thanks for this suggestion. Conducting the HDX-MS comparison between the WT and the D59C mutant is certainly interesting, especially given the growing amount of structural and biochemical/biophysical data available for this mutant. However, due to limited resources, we might consider doing it later.

(8) Have the authors considered utilising Li⁺ to infer how cation selectivity impacts the allostery? Do they expect similar stabilisation of a higher-affinity sugar binding state with all cations?

Thanks for this suggestion. We have demonstrated that Li⁺ also shows positive cooperativity with melibiose through ITC binding measurements. Li⁺ binds to MelB_{St} with higher affinity than Na⁺ but causes many different effects on MelB. It is worth investigating this thoroughly and individually. To address the second question, H⁺ is a poor coupling cation with minimal impact on melibiose binding. Since its pKa is around 6.5, only a small subpopulation of MelB_{St}

is protonated at pH 7.5. The order of sugar-binding cooperativity is the highest with Na^+ , followed by Li^+ and H^+ .

(9) MD of MelB suggests all transmembrane helices are reorientated during substrate translocation, yet substrate and cotransporter ligand binding only significantly impacts a small number of helices. Can the authors comment on the ensemble of states expected from each HDX experiment? The data presented here instead shows overall stabilisation of the transporter. This data can be compared to that of HDX on MFS sugar cation symporter XylE, where substrate binding induces a transition to OF state. There is no discussion of how this HDX data compares to previous MFS sugar transporter HDX. The manuscript could benefit from this comparison rather than a comparison to LacY. It is unlikely that there are universal mechanisms that can be inferred even from these model proteins. Highlighting differences instead between these transport systems provides broader insights into this protein class. Doi: 10.1021/jacs.2c06148 and 10.1038/s41467-018-06704-1.

The sugar translocation free-energy landscape simulations showed that both helix bundles move relative to the membrane plane. That analysis aimed to clarify a hypothesis in the field—that the MFS transporter can use an asymmetric mode to transition between inward- and outward-facing states. In the case of MelB, we clearly demonstrated that both domains move and each helix bundle moves as a unit, so the labeling changes were identified only in some extramembrane loops and a few highly flexible helices. Thanks for the suggestion about comparing with XylE. We will include a discussion on it.

(10) Additionally, the recent publication of SMFS data (by the authors: doi:10.1016/j.str.2022.11.011) states the following: "In the presence of either melibiose or a coupling Na^+ -cation, however, MelB increasingly populates the mechanically less stable state which shows a destabilized middle-loop C3." And "In the presence of both substrate and co-substrate, this mechanically less stable state of MelB is predominant." It would benefit the authors to comment on these data in contrast to the HDX obtained here. Additionally, is the C3 loop covered, and does it show the destabilization suggested by these studies? HDX can provide a plethora of results that are missing from the current analysis on ligand allostery. The authors instead chose to reference CD and thermal denaturation methods as comparisons.

Thank this reviewer for reading the single-molecule force spectroscopy (SMFS) study on MelB_{St}. The C3 loop mentioned in this SMFS article is partially covered in the dataset Mel or Mel plus Na^+ vs. Apo, and more coverage is in the Na^+ vs. Apo. In either condition, no deprotection was detected. Two possible reasons the HDX data did not reflect the deprotection are: 1) The changes were too subtle and did not pass the statistical tests and 2) the longest labeling time point was still insufficient to detect the changes; much longer labeling times should be considered in future studies.

Reviewer #3:

Summary:

The melibiose permease from *Salmonella enterica* serovar Typhimurium (MelB_{St}) is a member of the Major Facilitator Superfamily (MFS). It catalyzes the symport of a galactopyranoside with Na^+ , H^+ , or Li^+ , and serves as a prototype model system for investigating cation-coupled transport mechanisms. In cation-coupled symporters, a coupling cation typically moves down its electrochemical gradient to drive the uphill transport of a primary substrate; however, the precise role and molecular contribution of the cation in substrate binding and translocation remain unclear. In a prior study, the authors showed that the binding affinity for melibiose is increased in the presence of Na^+

by about 8-fold, but the molecular basis for the cooperative mechanism remains unclear. The objective of this study was to better understand the allosteric coupling between the Na^+ and melibiose binding sites. To verify the sugar-recognition specific determinants, the authors solved the outward-facing crystal structures of a uniport mutant D59C with four sugar ligands containing different numbers of monosaccharide units (α -NPG, melibiose, raffinose, or α -MG). The structure with α -NPG bound has improved resolution (2.7 Å) compared to a previously published structure and to those with other sugars. These structures show that the specificity is clearly directed toward the galactosyl moiety. However, the increased affinity for α -NPG involves its hydrophobic phenyl group, positioned at 4 Å-distance from the phenyl group of Tyr26 forms a strong stacking interaction. Moreover, a water molecule bound to OH-4 in the structure with α -NPG was proposed to contribute to the sugar recognition and appears on the pathway between the two specificity-determining pockets. Next, the authors analyzed by hydrogen-to-deuterium exchange coupled to mass spectrometry (HDX-MS) the changes in structural dynamics of the transporter induced by melibiose, Na^+ , or both. The data support the conclusion that the binding of the coupling cation at a remote location stabilizes the sugar-binding residues to switch to a higher-affinity state. Therefore, the coupling cation in this symporter was proposed to be an allosteric activator.

Strengths:

- (1) The manuscript is generally well written.
- (2) This study builds on the authors' accumulated knowledge of the melibiose permease and integrates structural and HDX-MS analyses to better understand the communication between the sodium ion and sugar binding sites. A high sequence coverage was obtained for the HDX-MS data (86-87%), which is high for a membrane protein.

Thank this reviewer for your positive comments.

Weaknesses:

- (1) I am not sure that the resolution of the structure (2.7 Å) is sufficiently high to unambiguously establish the presence of a water molecule bound to OH-4 of the α -NPG sugar. In Figure 2, the density for water 1 is not obvious to me, although it is indeed plausible that water mediates the interaction between OH4/OH6 and the residues Q372 and T373.

Thanks for your comments on the resolution. We will improve the density for the Water 1.

- (2) Site-directed mutagenesis could help strengthen the conclusions of the authors. Would the mutation(s) of Q372 and/or T373 support the water hypothesis by decreasing the affinity for sugars? Mutations of Thr 121, Arg 295, combined with functional and/or HDX-MS analyses, may also help support some of the claims of the authors regarding the allosteric communication between the two substrate-binding sites.

The authors thank this reviewer for the thoughtful suggestions. MelB_{St} has been subjected to Cys-scanning mutagenesis (<https://doi.org/10.1016/j.jbc.2021.101090>). Placing a Cys residue on the hydrogen bond-donor Q372 significantly decreased the transport initial rate, accumulation, and melibiose fermentation, with little effect on protein expression, as shown in Figure 2 of this *JBC* paper. Although no binding data are available, the poor initial rate of transport with a similar amount of protein expressed suggested that the binding affinity is apparently decreased, supporting the role of water-1 in the binding pocket for better binding. The T373C mutant retained most activities of the WT. We will discuss the functional characterizations of these two mutants. Thanks.

(3) The main conclusion of the authors is that the binding of the coupling cation stabilizes those dynamic sidechains in the sugar-binding pocket, leading to a high-affinity state. This is visible when comparing panels c and a from Figure S5. However, there is both increased protection (blue, near the sugar) and decreased protection in other areas (red). The latter was less commented, could the increased flexibility in these red regions facilitate the transition between inward- and outward-facing conformations?

Thanks for this important question. We will discuss the deprotected data in the conformational transition between inward-facing and outward-facing states. The two regions, loop8-9 and loop1-2, are located in the gate area on both sides of the membrane and showed increased deuterium uptakes upon binding of melibiose plus Na^+ . They are likely involved in this process.

The HDX changes induced by the different ligands were compared to the apo form (see Figure S5). It might be worth it for data presentation to also analyze the deuterium uptake difference by comparing the conditions sodium ion+melibiose vs melibiose alone. It would make the effect of Na^+ on the structural dynamics of the melibiose-bound transporter more visible. Similarly, the deuterium uptake difference between sodium ion+melibiose vs sodium ion alone could be analyzed too, in order to plot the effect of melibiose on the Na^+ -bound transporter.

We will analyze the data as suggested by this reviewer.

(4) For non-specialists, it would be beneficial to better introduce and explain the choice of using D59C for the structural analyses.

As response to the reviewer #1 at page 3, “Asp59 is the only site that responds to the binding of all coupling cations: Na^+ , Li^+ , or H^+ . Notably, this mutant selectively abolishes cation binding and cotransport. However, it still maintains intact sugar binding with slightly higher affinity and preserves the conformational transition, as demonstrated by an electroneutral transport reaction, the melibiose exchange, and fermentation assays with intact cells. Therefore, the structural data derived from this mutant are significant and offer important mechanistic insights into sugar transport. We will provide additional details during the revision.”.

(5) In Figure 5a, deuterium changes are plotted as a function of peptide ID number. It is hardly informative without making it clearer which regions it corresponds to. Only one peptide is indicated (213-226), I would recommend indicating more of them in areas where deuterium changes are substantial.

We appreciate this comment, which will make the plots more meaningful. In the previous article published in eLife (2024), we drew boxes to mark the transmembrane regions; however, it generated much confusion, such as why some helices are very short. The revised figure will label the full length of covered positions.

(6) From prior work of the authors, melibiose binding also substantially increases the affinity of the sodium ion. Can the authors interpret this observation based on the HDX data?

This is an intriguing mechanistic question. Based on current data, we believe that the bound melibiose physically prevents the release of Na^+ or Li^+ from the cation-binding pocket. The cation-binding pocket and surrounding regions, including the sugar-binding residue Asp124, show low HDX, supporting this idea. Since we lack a structure with both substrates bound,

figuring out the details structurally is challenging. However, we have a hypothesis about the intracellular Na^+ release as proposed in the 2024 JBC paper (<https://doi.org/10.1016/j.jbc.2024.107427>). After sugar release, the rotamer change of Asp55 will help Na^+ exit the cation pocket to the sugar pocket, and the negative membrane potential will facilitate the further movement from MelB to the cytosol. We will discuss this during the revision.

<https://doi.org/10.7554/eLife.108335.1.sa0>